Department of Computer Science and Engineering, UCE(A), OU

Course Title: Deep Learning B.E(CSE) VI-Sem - 2025-26
Time: 3 Hours
Maximum Marks: 30

Part A – Short Answer Questions (5 × 2 = 10 Marks)

Answer all questions.

1. Explain the concept of **Recurrent Neural Networks (RNNs)** and their key applications.
2. Differentiate between **GRU and LSTM** architectures.
3. What is an **attention mechanism**? Why is it important in sequence models?
4. Define **the concept of language models**?
5. What are **Generative Adversarial Networks (GANs)**? Mention their basic components.

Part B – Long Answer Questions (2 × 10 = 20 Marks)

Answer TWO questions. Each question has two parts (A & B).

6.

A) Explain the working of **Recurrent Neural Networks** and discuss the issue of vanishing gradients.
B) Describe **LSTM architecture** in detail with diagrams and equations.

7.

A) Explain different approaches to **object detection?**
B) Discuss **image augmentation techniques** and their importance in deep learning.

8.

A) Explain the architecture and training process of **Generative Adversarial Networks (GANs).**
B) Discuss tools used in deep learning such as **Jupyter Notebook, Google Colab, and AWS SageMaker,** highlighting their advantages.